Future & Promise

Async thumbnail then compose URL

Season 5 · Concurrency · Intermediate

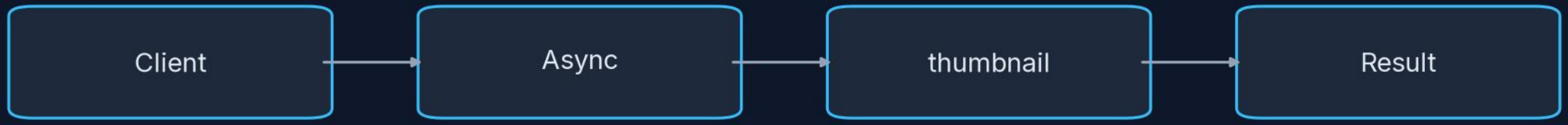
Today's Hook

- Problem: Async thumbnail then compose URL
- Naive pain cue: caught=id
- We will refactor to Future/Promise with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Async thumbnail then compose URL
- Constraints: solve «Async thumbnail then compose URL» with Future/Promise — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Async thumbnail then compose URL
- Success looks like: https://cdn/a.jpg

Structure



Future/Promise: Async thumbnail then compose URL

Real-World Scenario

- Running example: Async thumbnail then compose URL
- Key types: ThumbService
- Happy-path stdout cue: https://cdn/a.jpg
- Failure-path stdout cue: caught=id

Naive Code

```
package com.buildwithwaqar.patterns.concurrency.futurepromise.problem;

import java.util.*;

public final class NaiveThumb {
  public static String url(String id){ return "https://cdn/"+id+".jpg"; }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Async thumbnail then compose URL
- Without Future/Promise, call sites absorb the wrong details.
- Failure mode to show on camera: caught=id
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.concurrency.futurepromise.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.concurrent.locks.*;
import java.util.function.*;

final class ThumbService {
  CompletableFuture<String> render(String id){
    return CompletableFuture.supplyAsync(() -> {
      if (id==null || id.isBlank()) throw new IllegalArgumentException("id");
      return id+".bin";
    });
  }
  CompletableFuture<String> toUrl(String blob){ return CompletableFuture.completedFuture("https://cdn/"+blob.replace(".bin",".jpg")); }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.concurrency.futurepromise.problem;
import java.util.*; import java.util.concurrent.*; import java.util.concurrent.atomic.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Future/Promise: async thumbnail URL ===");
    System.out.println("--- naive ---"); System.out.println(NaiveThumb.url("a"));
System.out.println("--- future ---");
var s=new ThumbService();
System.out.println(s.render("vid").thenCompose(s::toUrl).join());
try { s.render("").join(); } catch(CompletionException e){ System.out.println("caught="+e.getCause().getMessage()); }
  }
}
```

Walkthrough Output

```
=== Future/Promise: async thumbnail URL ===
--- naive ---
https://cdn/a.jpg
--- future ---
https://cdn/vid.jpg
caught=id
```

Step-by-Step Trace

- Match stdout to stages — look for: https://cdn/a.jpg
- Then the injected failure: caught=id
- Re-run: java -cp … 38-future-promise ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
package com.buildwithwaqar.patterns.concurrency.futurepromise.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.concurrent.locks.*;
import java.util.function.*;

final class ThumbService {
  CompletableFuture<String> render(String id){
    return CompletableFuture.supplyAsync(() -> {
      if (id==null || id.isBlank()) throw new IllegalArgumentException("id");
      return id+".bin";
    });
  }
  CompletableFuture<String> toUrl(String blob){ return CompletableFuture.completedFuture("https://cdn/"+blob.replace(".bin",".jpg")); }
}
```

Trade-offs

- Use Future/Promise when «Async thumbnail then compose URL» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Future/Promise in product language.

Related Patterns

- Name the pattern usually confused with Future/Promise and separate the forces.
- Sketch the roles for «Async thumbnail then compose URL» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Async thumbnail then compose URL
- Naive pain → Future/Promise boundary → Demo stdout.
- Remember the failure cue: caught=id
- Break a ProblemTest assertion, watch it fail, restore it.